higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N5

30 July 2026

This marking guideline consists of 10 pages.

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N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
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QUESTION 1

1.1 1.1.1

$$y=\lim_{x\to 2}\left(\frac{1}{x-2}-\frac{1}{x^2-3x+2}\right) \qquad :(\infty-\infty)$$

$$=\lim_{x\to 2}\left(\frac{1}{x-2}-\frac{1}{(x-1)(x-2)}\right)$$

$$=\lim_{x\to 2}\left(\frac{x-1-1}{(x-2)(x-1)}\right)$$

$$\lim_{x\to 2}\left(\frac{x-2}{(x-2)(x-1)}\right)$$

$$=1$$ (2)

1.1.2

$$y=\lim_{x\to\infty}\frac{\ln(x+2)}{\log_2 x} \qquad :\frac{\infty}{\infty}$$

$$=\frac{\lim\limits_{x\to\infty}\left(\frac{1}{x+2}\right)}{\left(\frac{1}{x\ln 2}\right)}=\lim_{x\to\infty}\frac{x\ln 2}{x+2} \qquad :\frac{\infty}{\infty}$$

$$=\lim_{x\to\infty}\frac{\ln 2}{1}$$

$$=\ln 2 \quad \text{OR} \quad 0{,}693$$ (3)

1.2

$$f(x)=\frac{x^2+5x-24}{x^2-x-6}$$

$x=-2$ *OF* $x=3$ (2)

[7]

QUESTION 2

2.1

$$f(x+h)=\frac{1}{4}x^{-4}-x^{-5}h+\frac{5}{2}x^{-6}h^2+\ldots$$

$$f'(x)=\lim_{h\to 0}\frac{\frac{1}{4}x^{-4}-x^{-5}h+\frac{5}{2}x^{-6}h^2+\ldots-\frac{1}{4}x^{-4}}{h}$$

$$=\lim_{h\to 0}\frac{-x^{-5}h+\frac{5}{2}x^{-6}h^2+\ldots}{h}$$

(5)

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$$= \lim_{h \to 0} -x^{-5} + \frac{5}{2}x^{-6}h + \ldots$$

$$= -x^{-5}$$

2.2 2.2.1 $y = cosec^4(\tan 2x)$

$$\frac{dy}{dx}$$
$$= 4\,cosec^3(\tan 2x) \times \left[-cosec(\tan 2x)\cot(\tan 2x)\right] \times sec^2(\tan 2x) \times 2$$ (4)

2.2.2 $$y = \frac{\sqrt[3]{x} - 2xe^x}{3x+1}$$

$$\frac{dy}{dx} = \frac{(3x+1)\left[\frac{1}{3}x^{-\frac{2}{3}} - 2\left(e^x + xe^x\right)\right] - 3\left(\sqrt[3]{x} - 2xe^x\right)}{(3x+1)^2}$$ (4)

2.3 $y = (sec\,3x)^{4x^2}$

$$\ln y = 4x^2 \ln(sec\,3x)$$

$$\frac{1}{y}\frac{dy}{dx} = 8x\ln(sec\,3x) + 4x^2 \times \frac{1}{sec\,3x} \times sec\,3x\tan 3x \times 3$$

$$\frac{dy}{dx} = y\left[8x\ln(sec\,3x) - 12x^2\tan 3x\right]$$

$$= (sec\,3x)^{4x^2}\left[8x\ln(sec\,3x) - 12x^2\tan 3x\right]$$ (4)

2.4 $sec(x-y) = y\,cosec\,x$

$$sec(x-y)\tan(x-y)\left(1 - \frac{dy}{dx}\right) = cosec\,x\frac{dy}{dx} - cosec\,x\cot x$$

$$sec(x-y)\tan(x-y) - sec(x-y)\tan(x-y)\frac{dy}{dx} = cosec\,x\frac{dy}{dx} - cosec\,x\cot x$$

$$\left[sec(x-y)\tan(x-y) + cosec\,x\right]\frac{dy}{dx} = sec(x-y)\tan(x-y) + cosec\,x\cot x$$

$$\frac{dy}{dx} = \frac{sec(x-y)\tan(x-y) + cosec\,x\cot x}{sec(x-y)\tan(x-y) + cosec\,x}$$ (5)

[22]

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QUESTION 3

3.1 3.1.1 $f(x)=3x^3-4x+3$

$f'(x)=9x^2-4=0$

$(3x+2)(3x-2)=0$

$x=-\frac{2}{3}$ OR $x=\frac{2}{3}$

$y=1,222$ $y=4,778$

Thus, the coordinates of the turning points are $\left(\frac{2}{3};\ 1,222\right)$ and $\left(-\frac{2}{3};4,778\right)$ (3)

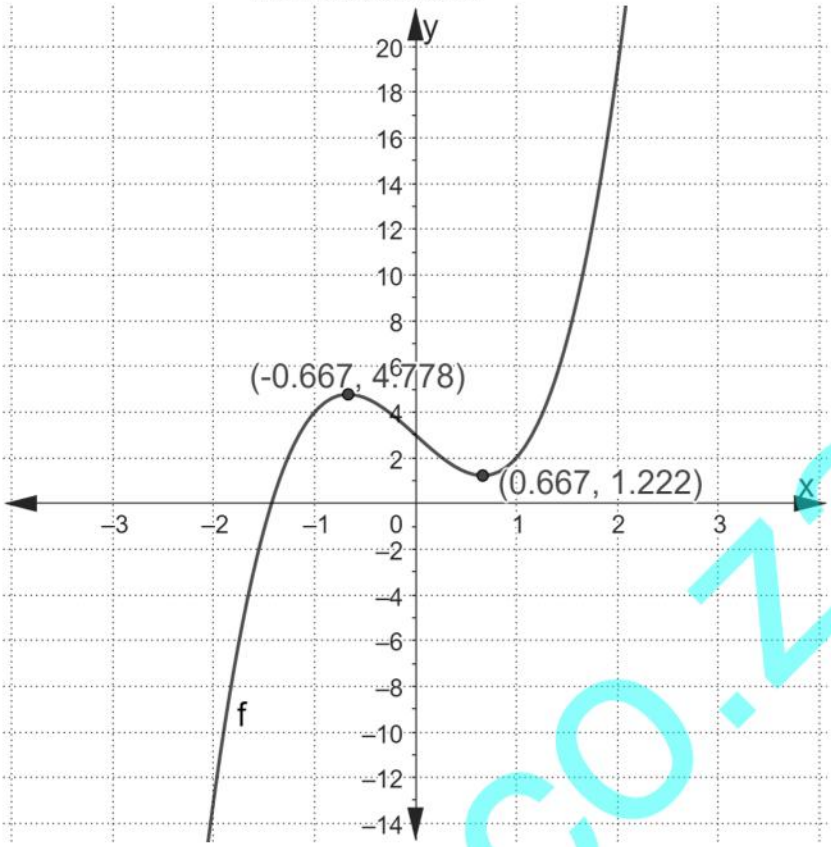
3.1.2

x	-2	-1	0	1	2
y	-13	4	3	2	19

= 1 mark for every 2 correct answers. (2)

3.1.3 (2)

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1 mark for the shape.
1 mark for the indication of the turnings points on the graph.

3.1.4 *Let* $x_0 = -1,3$

$f(-1,3) = 1,609$

$f'(-1,3) = 11,21$

$x_1 = -1,3 - \frac{1,609}{11,21}$

$= -1,444$

$x_2 = -1,444 - \frac{-0,256}{14,74}$

$x_2 = -1,427$ (4)

3.2 $P = 2x + y$

$100 = 2x + y$

$y = 100 - 2x$

$A = xy$

(6)

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$A = x(100 - 2x)$

$A = 100x - 2x^2$

$A' = 100 - 4x = 0$

$4x = 100$

$x = 25\,m$

$y = 50\,m$

Largest area

$A = xy = 25 \times 50 = 1250\ \ m^2$

3.3 $h = 2 + 24{,}5t - 4{,}9t^2$

$h' = 24{,}5 - 9{,}8t = 0$

$9{,}8t = 24{,}5$

$t = 2{,}5\,s$ (3)

[20]

QUESTION 4

4.1 4.1.1 $I = \int \frac{\tan x \sec^2 x}{1 - \tan^2 x}\,dx$

	ALTERNATIVE WORKINGS
$let\ \ u = 1 - \tan^2 x$	$let\ \ f(x) = 1 - \tan^2 x$
$-\frac{1}{2}du = \tan x \sec^2 x\,dx$	$f'(x) = -2\tan x \sec^2 x\,dx$
$I = -\frac{1}{2}\int \frac{1}{u}du$	$= -\frac{1}{2}\int \frac{-2 \times \tan x\, se}{1 - \tan^2 x}$
$= -\frac{1}{2}\left[\ln u\right] + C$	$= -\frac{1}{2}\ln(1 - \tan^2$
$= -\frac{1}{2}\ln(1 - \tan^2 x) + C$	

(3)

4.1.2 $\int \cos^2 6x\,dx$

(3)

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$$=\frac{1}{2}\int(1+\cos 12x)\,dx$$

$$=\frac{1}{2}x+\frac{1}{24}\sin 12x\square+C$$

4.1.3 $$\int \sin(3x)\sin\left(\frac{4}{3}x\right)dx$$

$$=\int\frac{1}{2}\left[\cos\frac{5}{3}x+\cos\frac{13}{3}x\right]dx$$

$$=\frac{1}{2}\left[\frac{3}{5}\sin\frac{5}{3}x\ +\frac{3}{13}\sin\frac{13}{3}x\ \right]+C$$

$$=\frac{3}{10}\sin\frac{5}{3}x+\frac{3}{26}\sin\frac{13}{3}x+C$$ (3)

4.1.4 $$\int\frac{x^3-3x+2}{x+1}dx$$

$$=\int\left(x^2-x-2+\frac{4}{x+1}\right)dx$$

$$=\frac{1}{3}x^3-\frac{1}{2}x^2-2x+4\ln(x+1)+C$$ (5)

4.1.5 $$\int 2\sin x\ln(sec\,x)\,dx$$

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let $u\ \vee\ f(x)=\ln(sec\,x)$ $\qquad dv\ \vee\ g'(x)=\sin x\,dx$

$du\ \vee\ f'(x)=\tan x\,dx$ $\qquad v\ \vee\ g(x)=-\cos x$

$$\int 2\sin x\ln(sec\,x)\,dx=2\left[-\cos x\ln(sec\,x)-\int-\cos x\tan x\,dx\right]$$

$$=2\left[-\cos x\ln(sec\,x)+\int\sin x\,dx\right]$$

$$=-2\cos x\ln(sec\,x)-2\cos x+C$$ (4)

4.2 $$\int\frac{x-7}{x^2+x-2}dx$$

$$\frac{x-7}{(x+2)(x-1)}=\frac{A}{x+2}+\frac{B}{x-1}$$

$$x-7=A(x-1)+B(x+2)$$

---

let $x=1;$ $\qquad B=-2$ (5)

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let $x=-2$; $A=3$

$$\int\left(\frac{3}{x+2}-\frac{2}{x-1}\right)dx$$

$$=3\ln(x+2)-2\ln(x-1)+C$$

[23]

QUESTION 5

5.1
$$\int_0^{\frac{\pi}{4}} 5x\sin 2x\,dx$$

$$=5\int_0^{\frac{\pi}{4}} x\sin 2x\,dx$$

$$=5\left\{\left[-x\frac{\cos 2x}{2}\right]_0^{\frac{\pi}{4}}+\int_0^{\frac{\pi}{4}}\frac{\cos 2x}{2}dx\right\}$$

$$=5\left\{\left[-x\frac{\cos 2x}{2}+\frac{\sin 2x}{4}\right]_0^{\frac{\pi}{4}}\right\}$$

$$=\frac{5}{4}\quad \text{OR}\quad 1{,}25$$

(4)

5.2 5.2.1 $2x=-x^2+6x$

(3)

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$x^2 - 4x = 0$

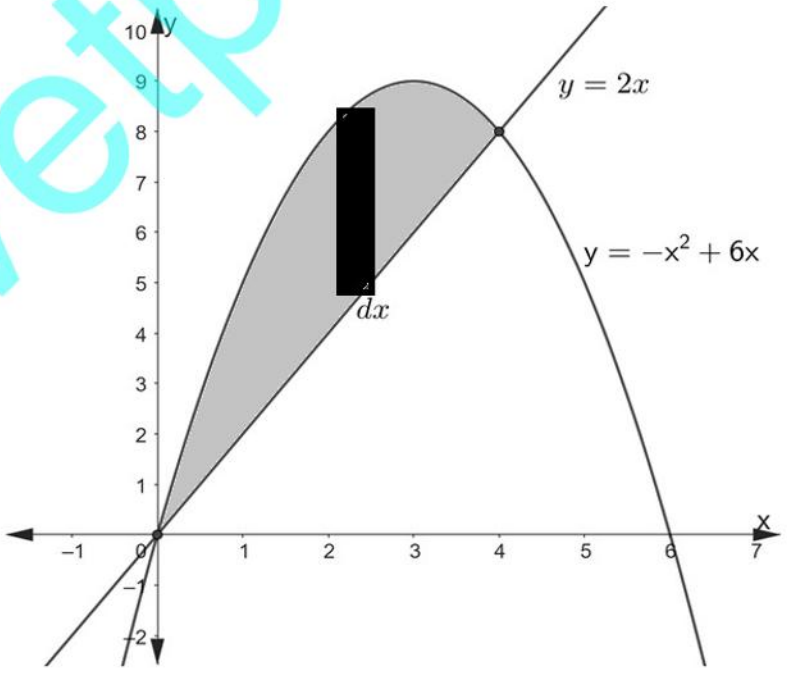
$x(x-4) = 0$

$x = 0 \quad \vee \quad x = 4$

$y = 0 \quad \vee \quad y = 8$

Thus, the coordinates of the points of intersection are $(0;\ 0)$ and $(4;8)$.

5.2.2



= 1 mark for indication of each enclosed area
= 1 mark for indication of the vertical strip (2)

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5.2.3

$$A=\int_a^b (y_1-y_2)\,dx$$

$$=\int_0^4 \left[(-x^2+6x)-2x\right]dx$$

$$=\int_0^4 (-x^2+4x)\,dx$$

$$=\left[-\frac{1}{3}x^3+2x^2\right]_0^4$$

$$=10{,}667 \ \ units^2 \ \text{ OR } \ \frac{32}{3}$$

(3)

5.2.4

$$V=\pi\int_a^b (y_1^{\,2}-y_2^{\,2})\,dx$$

$$=\pi\int_0^4 \left[(-x^2+6x)^2-(2x)^2\right]dx$$

$$=\pi\int_0^4 (x^4-12x^3+32x^2)\,dx$$

$$=\pi\left[\frac{1}{5}x^5-3x^4+\frac{32}{3}x^3\right]_0^4$$

$$=\frac{1792\pi}{15}=375{,}316 \ \ units^3$$

(5)

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5.3 $I_z = \frac{1}{2} m R^2$

$= \frac{1}{2} \times 30(1.2)^2$

$= 21,6 \ kg.m^2$ (3)

[20]

QUESTION 6

6.1 $\frac{dy}{dx} = x e^{2x^2}(1+y)$

$\frac{1}{1+y} dx = x e^{2x^2} dx$

$\ln(1+y) = \frac{1}{4} e^{2x^2} + C$ (3)

6.2 $\frac{d^2y}{dx^2} = 6x + 8\cos 2x$, *with* $\frac{dy}{dx} = 3$ and $y = 2$ *at* $x = 0$

$\frac{dy}{dx} = 3x^2 + 4\sin 2x + A$

$3 = 3(0)^2 + 4\sin 2(0) + A$

$A = 3$

$\frac{dy}{dx} = x^3 + 4\sin 2x + 3$

$y = x^3 - 2\cos 2x + 3x + B$

$2 = 0^3 - 2\cos 2(0) + 3(0) + B$

$B = 4$

$y = x^3 - 2\cos 2x + 3x + 4$ (5)

[8]

TOTAL: 100

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